BMI5206 Group 5

"Digital Nun": EHR Mobile Application

Context of our project:

- Our project is about a patient-centred mobile application that enables medical providers, patients and their caretakers to access and aggregate health information across medical institutions, upload pictures or PDF documents of health documents, and upload home health monitoring data.

Persona 1

- Name: Mr Tan
- Age 62
- Occupation: Retired
- Tech savviness: Low to medium
- Context: Mr. Tan is a 62 year old patient with common chronic diseases in older adults. He has diabetes and high blood pressure. He visits different hospitals and clinics. His medical records are spread across many systems and paper reports. He can take photos with his phone and use simple app functions, but he cannot manage complex digital tools. He does not understand medical terms or lab numbers well. When he receives test results, he often feels confused and anxious. He relies on his daughter to help explain his condition and medical history.

Persona 2

- Name: Miss Tan
- Age: 32
- Occupation: Marketing Manager
- Tech savviness: High
- Context: Daughter of Mr Tan. Ms. Tan is a working adult and the main caregiver for her father. She helps manage his medical information across different healthcare providers. She collects paper reports, photos, and PDFs from many sources. She often needs to explain her father’s chronic conditions and other health problems to different doctors again and again. This takes time and increases mental stress. She worries about missing important information and wants a clear way to organize records and track health changes over time.

Persona 1 Scenario

- **Persona is doing**: Mr. Tan has just received a lab report from his recent hospital visit and feels anxious because he does not understand the results.
- **Takes/Opens**: He takes out his smartphone and opens the patient-centered health app.
- **Clicks on**: He clicks on the “Upload Document” button to take a photo of his lab report. After uploading, he clicks on “View Summary” where the app shows simplified explanations of his blood sugar and blood pressure readings.
- **Persona is doing**: He reviews the color-coded charts that indicate which values are within a healthy range.
- **Clicks on**: He clicks the “Share with Caregiver” option to notify his daughter that the report is available for review.

Persona 2 Scenario

- **Persona is doing**: Ms. Tan receives a notification from the app that her father's lab report has been uploaded. She wants to make sure all his recent medical records are organized.
- **Takes/Opens**: She opens the app on her tablet and navigates to her father's "Health Timeline" view.
- **Clicks on**: She clicks on the newly uploaded lab report to view the details and simplified explanations. She then clicks on "Add Notes" to include her observations and reminders from the doctor.
- **Persona is doing**: She reviews past reports and compares them with the new results to track trends in her father's blood sugar and blood pressure.
- **Clicks on**: She clicks on "Share with Doctor" to securely send a complete set of recent records before her father's next consultation.

Best ways to perform user research

Contextual Inquiry and Observation

- What it is: Observing users in their natural environment while they perform tasks related to health management.
- Why it works:
  - For Mr. Tan, you can see how he handles medical documents, uses his phone, and interacts with healthcare staff.
  - For Ms. Tan, you can understand the stress points of gathering documents, explaining health history to providers, and managing schedules.
- How to do it:
  - Visit patients at home or accompany caregivers during hospital visits.
  - Take notes on how documents are handled, when confusion arises, and what workarounds they use.

Semi-Structured Interviews

- What it is: Guided conversations with key questions that are left flexible to explore unexpected issues.
- Why it works:
  - Captures emotional context (anxiety, frustration) and unmet needs.
  - Allows patients to express what is hard for them without overwhelming them with unfamiliar jargon.
- How to do it:
  - Ask about daily routines, document handling, interactions with multiple providers, and use of technology.
  - For Ms. Tan, focus on workflow challenges and the cognitive load of managing her father's health.

Self-Reporting

- What it is: Users log their interactions and experiences over a period of time.
- Why it works:
  - Chronic disease management is ongoing; snapshots from one visit may miss critical issues.
  - Helps track patterns like missed appointments, forgotten measurements, or repeated explanations to doctors.
- How to do it:
  - Mr. Tan can take quick notes or photos of reports at home.
  - Ms. Tan can keep a short daily log of tasks, challenges, and communication with healthcare providers.

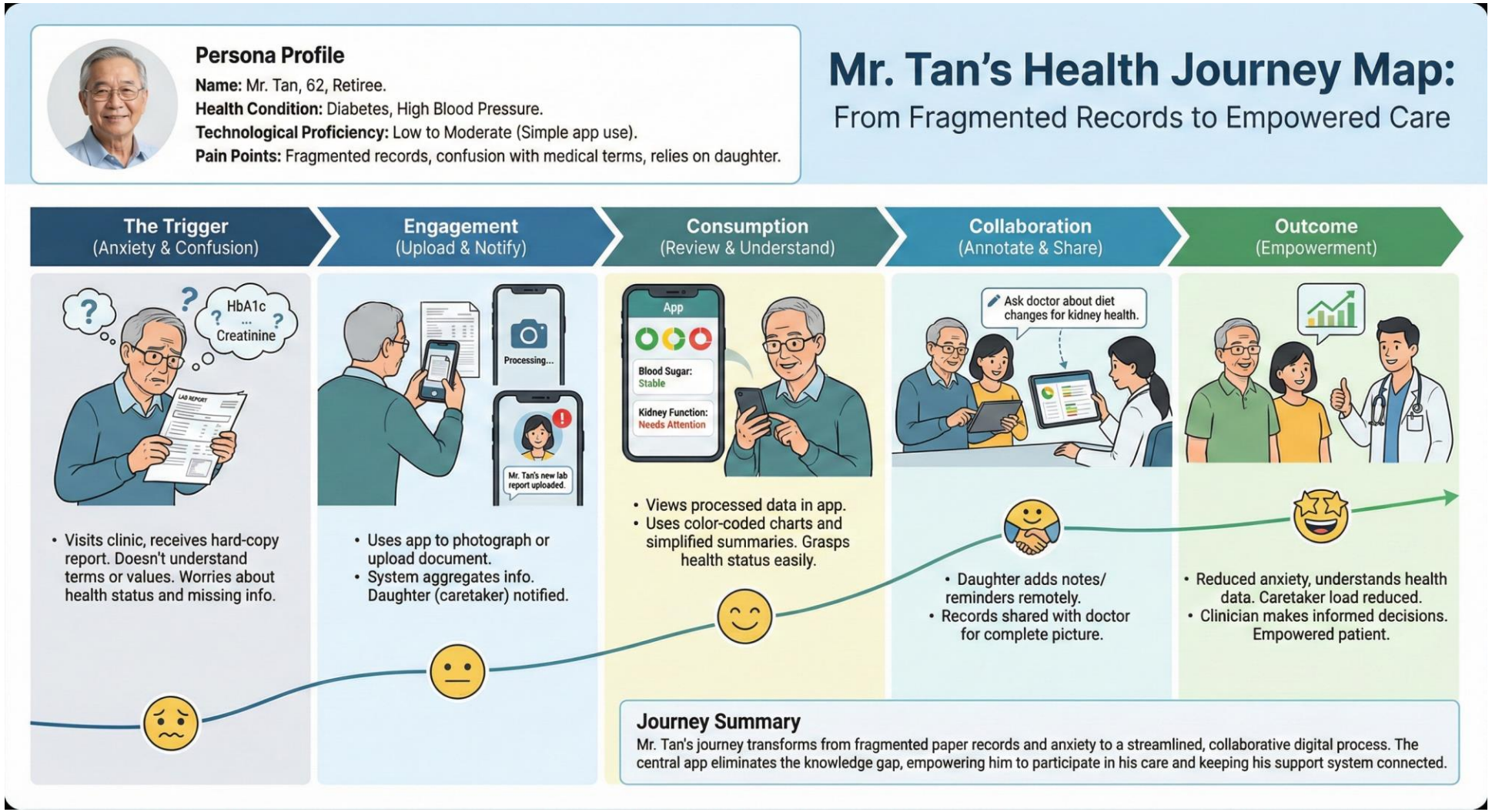
Usability Testing with Low-Fidelity Prototypes

- What it is: Test app mockups with users performing realistic tasks.
- Why it works:
  - Evaluates whether Mr. Tan can navigate simple functions (uploading photos, viewing summaries).
  - Tests if Ms. Tan can efficiently organize, annotate, and share multiple records.
- How to do it:
  - Use paper prototypes or simple interactive digital prototypes.
  - Give task scenarios like “Upload a lab report and share it with your caregiver” or “Compare last week’s readings with today.”

Participatory Design Workshops

- What it is: Involve users in designing solutions rather than just observing.
- Why it works:
  - Leverages Ms. Tan's expertise in managing her father's records and Mr. Tan's perspective as a patient.
  - Generates ideas that are realistic for their actual routines and abilities.
- How to do it:
  - Conduct small sessions with paper sketches or post-its to prioritize features (e.g., document aggregation, home monitoring uploads).

User Journey



Design Metaphor: The Digital Nun

Stewardship of Care for Elderly Patients (e.g., Mr. Tan) – Reducing Cognitive Load, Ensuring Comfort

1. The Scribe
(Aggregating Records)

Historical Role: Maintaining continuity of hospital records.

App Function: Photo upload of notes/reports. No sorting required.

Design Interaction: Quiet acceptance. Confirms filing. Handles backend.

Benefit: Unified keeper of the ledger. Removes organizational burden.

2. The Translator
(Contextualized Education)

Historical Role: Translating complex doctor orders.

App Function: Explains confusing lab results.

Design Interaction: Gentle interpreter using plain language, not raw data.

Benefit: Ensures personal understanding of health data.

3. The Vigilant Watcher
(Safe & Timely Monitoring)

Historical Role: Bedside presence watching for danger.

App Function: Silent monitoring of vitals trends.

Design Interaction: Only speaks up for concerning trends with a tone of concern, avoiding alert fatigue.

Benefit: Eliminates anxiety of missing information, prevents panic.

4. The Humble Interface
(Equitable & Accessible)

Historical Role: Serving all with simple resources.

App Function: UI for low tech savviness.

Design Interaction: High contrast, warm colors, large fonts. Gentle error recovery ("Did you mean...?").

Benefit: Supports varying digital literacy, minimizes mental burden.

Summary for Mr. Tan's Journey

- Mr. Tan is a charge under care, not just a user.
- **When anxious:** Simplifies view to essentials.
- **When confused:** Explains terms in his language.
- **At doctor visit:** Hands over complete "chart" for evidence-based decisions.